Lesson 15: Rational & Reciprocal Functions — Practice Worksheet

Name: ______________________________ Date: ____________

Attempt every question. Show your working. The sections increase in difficulty.

Section 1 — Warm-up (foundation)

1. Vertical asymptote of $y = 1/(x - 4)$?

2. Vertical asymptote of $y = 1/(x + 6)$?

3. Horizontal asymptote of $y = 1/x + 5$?

4. Domain of $y = 1/(x - 2)$?

5. Horizontal asymptote of $y = 1/x$?

6. Vertical asymptote of $y = 1/x$?

Section 2 — Core practice

7. Asymptotes of $y = (2x + 1)/(x - 5)$.

8. Asymptotes of $y = (3x - 2)/(x + 1)$.

9. Horizontal asymptote of $y = (5x)/(x - 2)$?

10. Vertical asymptote of $y = (x + 3)/(2x - 6)$?

11. Domain of $y = (x + 1)/(x - 7)$?

12. Horizontal asymptote of $y = (4x + 1)/(2x - 3)$?

Section 3 — Challenge & exam-style

13. State the asymptotes, domain and range of $y = (x - 1)/(x + 2)$.

14. A curve $y = a/(x - 3) + 2$ has a horizontal asymptote. State it.

15. For $y = (6x + 5)/(3x - 9)$, find both asymptotes.

16. Explain why $y = 1/x$ never takes the value 0.